

MTH203: Multivariate Calculus

Monsoon 2025

Name: _____

Roll No.: _____

Tutorial Quiz 5

Question:

1. Evaluate the triple integral

$$\int_0^1 \int_0^1 \int_0^1 (x^2 + y^2 + z^2) dz dy dx$$

(1)

2. Find the volume of the region enclosed by the cylinder $x^2 + y^2 = 4$, the plane $z = 0$, and the plane $y + z = 4$.

(2)

Please note: There is no partial marking. Only integer marking.